

Initial Proposal - Group 40

- **System Description:**

The system chosen for this project is P.I.T.C.H., a microservice-based web application developed as part of our capstone project. P.I.T.C.H. is an AI-assisted platform that helps users improve and practice their pitches and presentations. The system has a Next.js web frontend and a NestJS microservices backend behind an API gateway. Services communicate via HTTP APIs and RabbitMQ for async workflows, and each service owns its data with Redis for caching. These services handle core functionality such as request processing, data management, and AI-related tasks.

The project will evaluate performance characteristics such as response time, throughput, and scalability when the system is subjected to different levels of load.

- **Research Questions:**

This project aims to answer the following questions:

1. How do response time and throughput change as the number of concurrent users increase?
2. Which backend services become performance bottlenecks under heavy load?
3. How does scaling backend services affect overall system performance?

- **Background & Motivation:**

Microservice architectures are now common in modern cloud-based systems because they allow applications to scale and evolve more easily. At the same time, spreading functionality across multiple services can introduce performance challenges, especially as traffic increases. Latency, resource contention, and service bottlenecks often only become visible under realistic workloads.

Studying the performance of a real microservice application like our capstone project helps highlight these challenges in a practical setting. The results of this evaluation are relevant to both industry and research, as performance analysis and scalability testing are essential for ensuring reliable user experiences in modern web and AI-driven applications.